

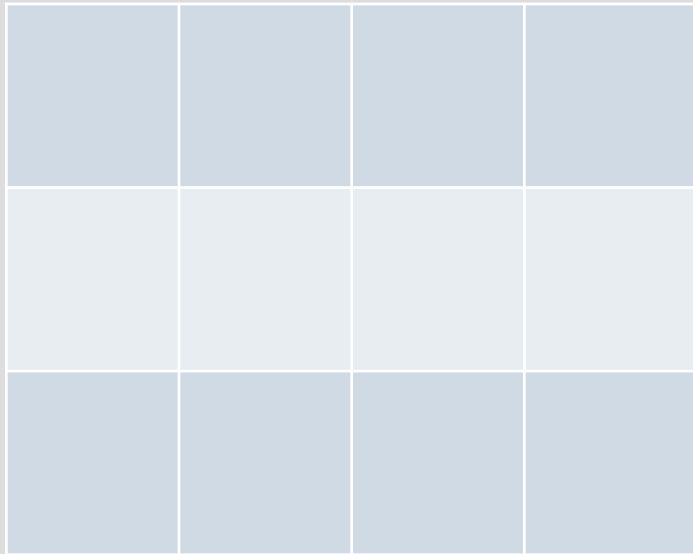
COUNTING

CSI 219 – Discrete Mathematics

THE PIGEONHOLE PRINCIPLE

HOW TO PLACE THESE PIGEONS?

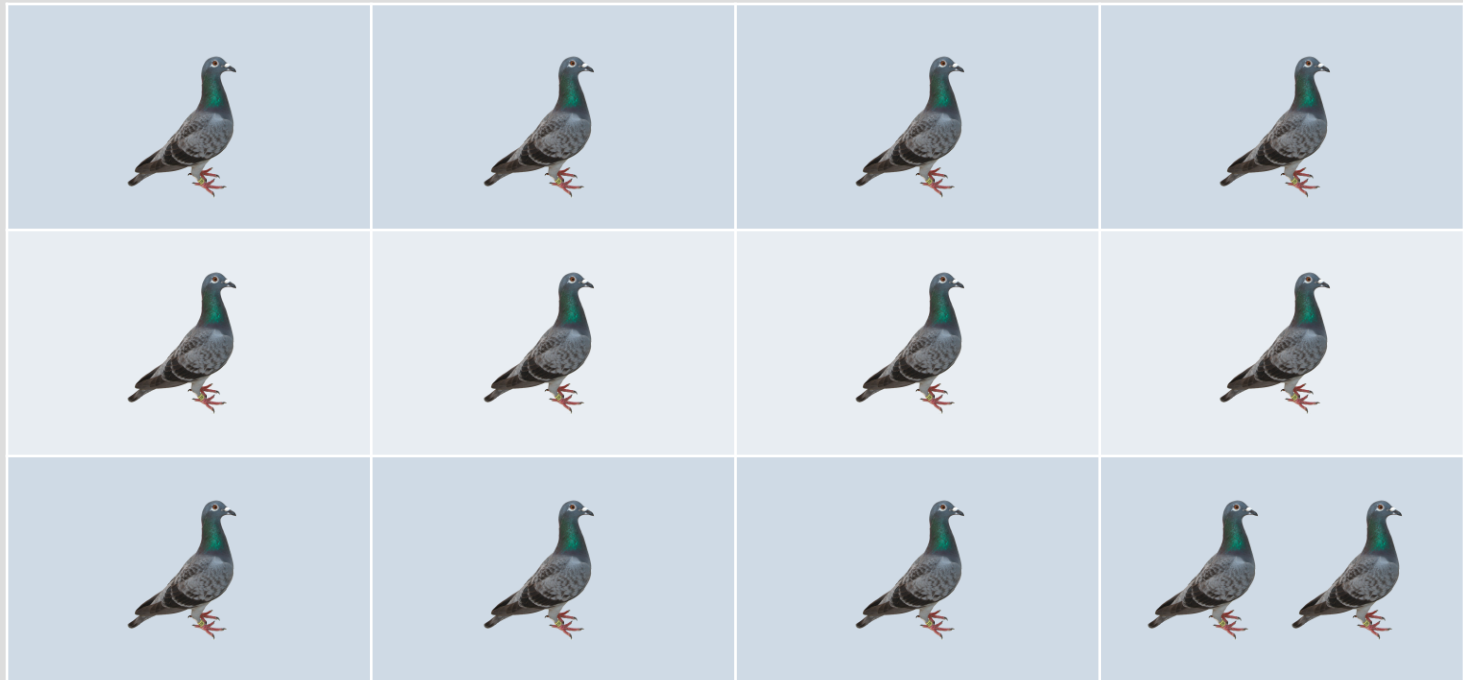
12 PIGEONHOLES



13 PIGEONS



HOW TO PLACE THESE PIGEONS?



THE PIGEONHOLE PRINCIPLE

- If k is a positive integer, and $k + 1$ or more objects are placed into k boxes, then there is at least one box that contains two or more objects

EXAMPLE

- How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

GENERALIZED PIGEONHOLE PRINCIPLE

- If N objects are placed into k boxes, then there is at least one box containing at least $\left\lceil \frac{N}{k} \right\rceil$ objects

EXAMPLE

Among 100 people, at least how many are guaranteed to be born in the same month?

$$\text{Answer: } \left\lceil \frac{100}{12} \right\rceil = 9$$

EXAMPLE

- What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

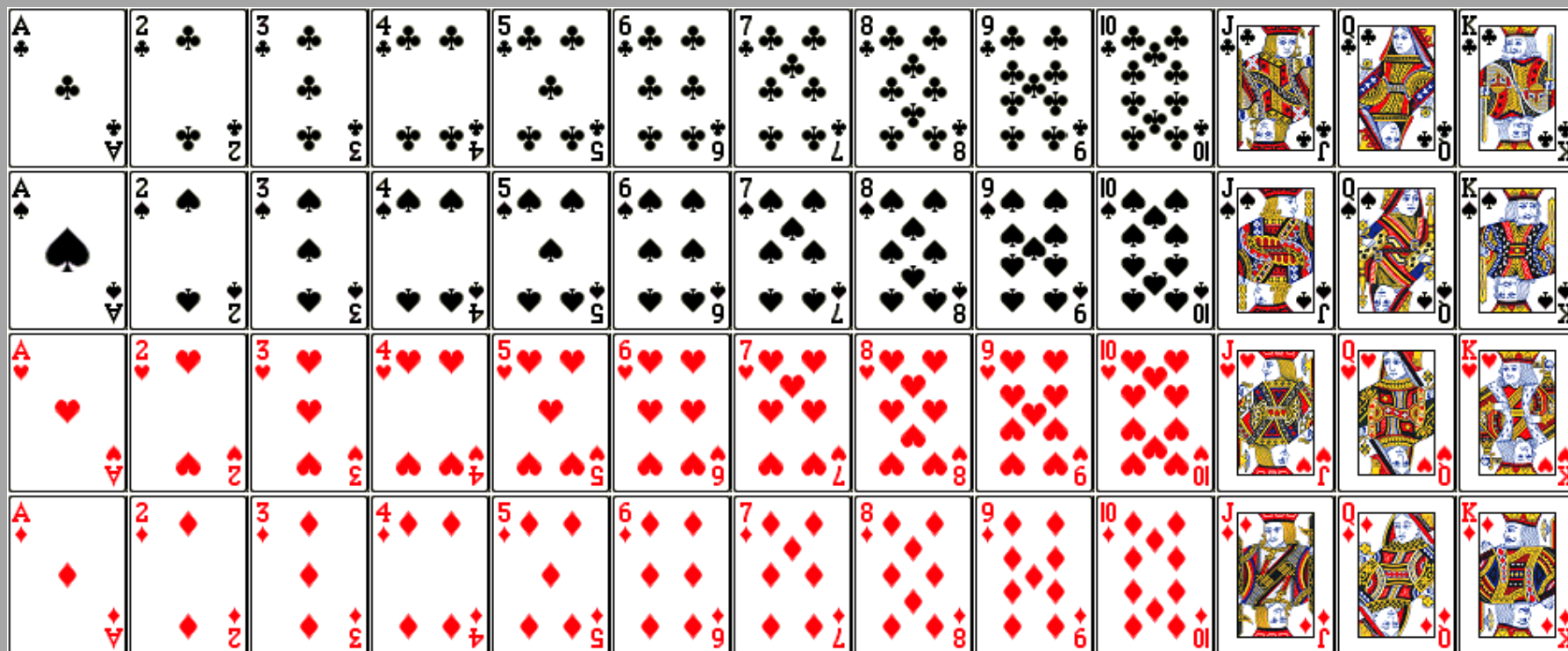
EXERCISE

How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

CARDS TUTORIAL

- There are four suits
 - Spades
 - Hearts
 - Diamonds
 - Clubs
- 13 cards of each suit
- Total $4 \times 13 = 52$ cards





EXERCISE

- a)** How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?
- b)** How many must be selected to guarantee that at least three hearts are selected?

EXERCISE

A drawer contains a dozen brown socks and a dozen black socks, all unmatched. A man takes socks out at random in the dark.

- a) How many socks must he take out to be sure that he has at least two socks of the same color?
- b) How many socks must he take out to be sure that he has at least two black socks?

EXERCISE

Telephone number format: $NXX-NXX-XXXX$,
where $N = 2 \text{ to } 9$, $X = 0 \text{ to } 9$

The first three digits form the area code

25 million people in the state

At least how many area codes needed?